

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMAL PROPERTIES OF MATTER

LECTURE-01

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1** Heat
- 2** Temperature
- 3** Thermal Expansion
- 4** Thermal Expansion of Solids
- 5** Liquids and gases

Thermal Properties of Matter

Heat, temperature, thermal expansion; thermal expansion of solids, liquids and gases, anomalous expansion of water; specific heat capacity; C_p , C_v - calorimetry; change of State latent heat capacity. Heat transfer-conduction, convection and radiation, thermal conductivity, qualitative ideas of Blackbody radiation, Wein's displacement Law, Stefan's law.

Heat

[Q]

lecture no. 1

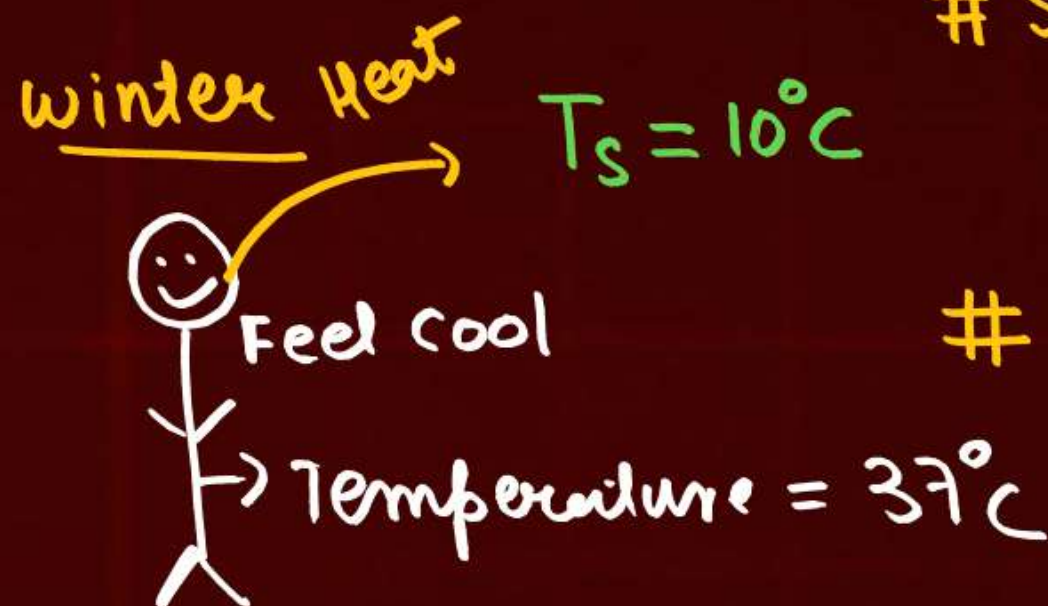
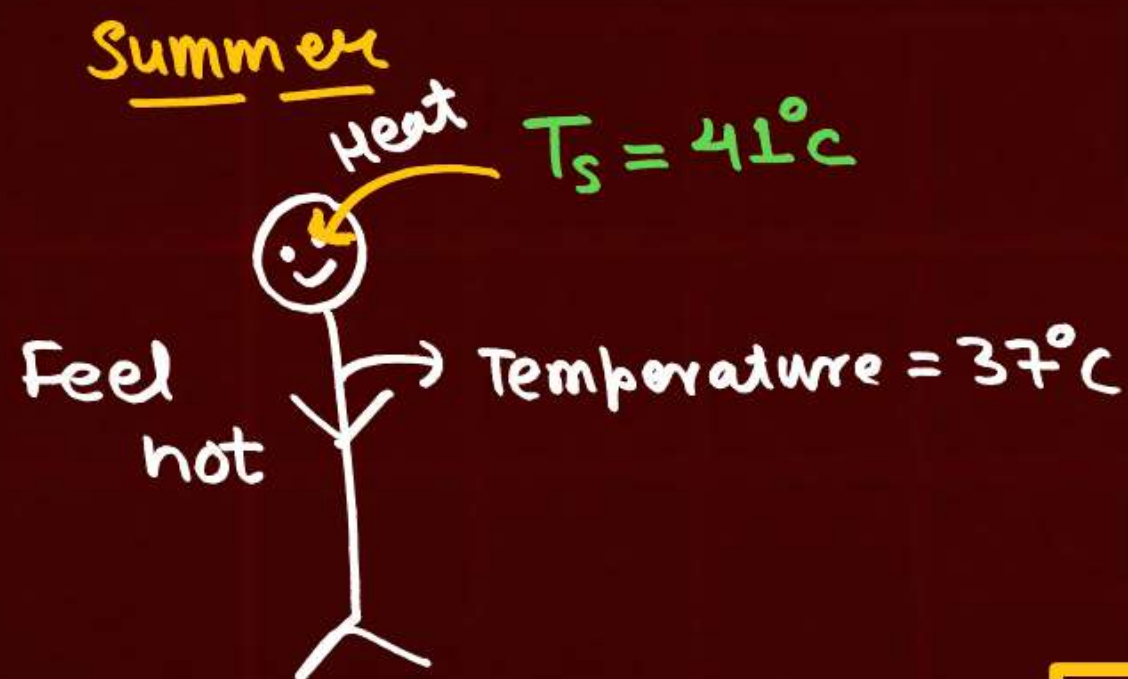


①

The form of **Energy** which produce the sensation of Warmness.

SI unit Heat
Joule

C.G.S unit of
heat
erg



Note :- Heat travel from **high temperature** to **low temperature**



Temperature

The degree of **hotness** or **coolness**

Body Temperature $\uparrow\uparrow$ **hot**

Body Temp $\downarrow\downarrow$ **cool**

unit of temperature ($^{\circ}\text{C}$, K, F)

$^{\circ}\text{C}$ $\left\{ \begin{array}{l} \rightarrow \text{K} \\ \rightarrow \text{F} \end{array} \right.$

$$\text{K} = ^{\circ}\text{C} + 273$$

$$\text{F} = \frac{9}{5}^{\circ}\text{C} + 32$$



Numerical.

$$\underline{\underline{Q}} \quad 10^{\circ}\text{C} = \underline{\underline{283}} \text{ K}$$

$$\underline{\underline{\text{Sol}}} \quad \underline{\underline{10^{\circ}\text{C}}}$$

$$\text{K} = ^{\circ}\text{C} + 273$$

$$= 10 + 273$$

$$= 283 \text{ K}$$

$$(a) 273 \text{ K}$$

$$(b) 283 \text{ K}$$

$$(c) 293 \text{ K}$$

$$(d) 393 \text{ K}$$

③

Q Convert $5^{\circ}\text{C} = \underline{41} \text{ F}$

④

Sol

$$F = \frac{9}{5}^{\circ}\text{C} + 32$$
$$= \frac{9}{\cancel{5}} \times \cancel{5} + 32$$

$$= 9 + 32$$

$$F = 41$$

(a) 41 F

(b) 42 F

(c) 43 F

(d) 45 F

Thermal Expansion

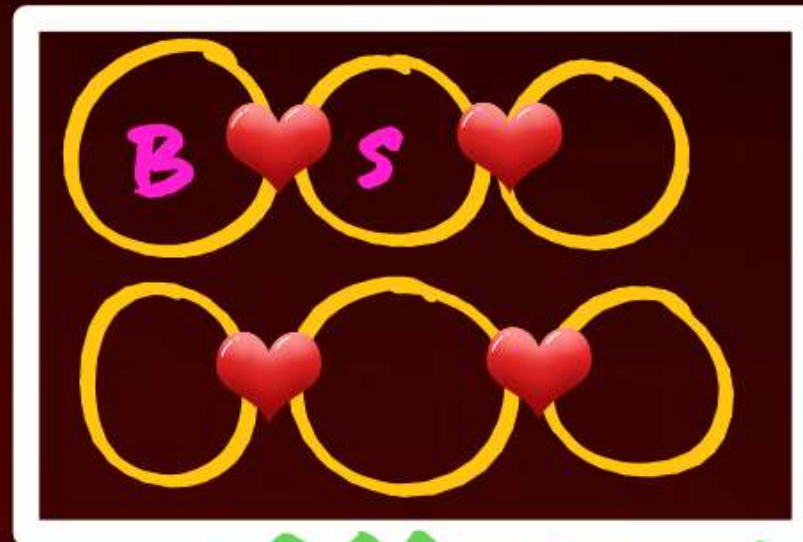
• when you give heat to the Body, its temperature increases

⇒ Intermolecular Force get weaker

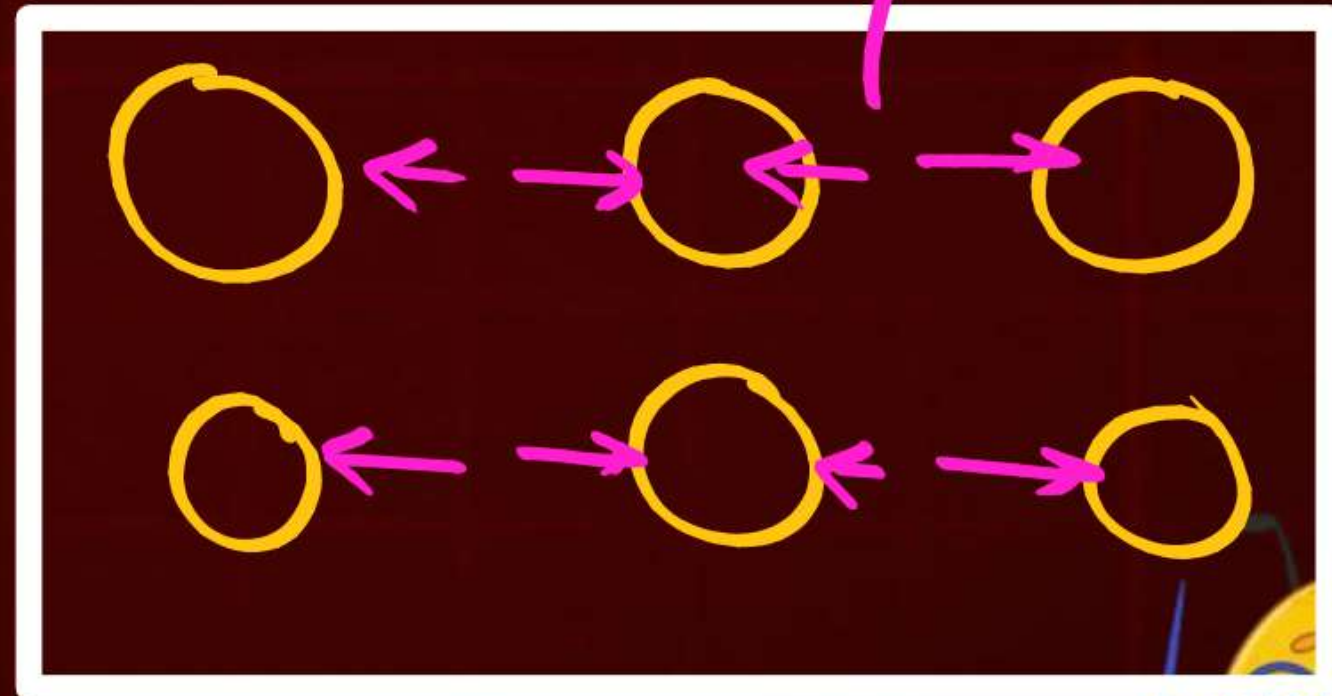
⇒ And Intermolecular distance increases

This is called Thermal

Expansion



heat ↑↑↑ Temp



Space ↑↑
Volume ↑↑



Thermal Expansion

⑥

Important

1 Solid



⇓ heat temperature ↑



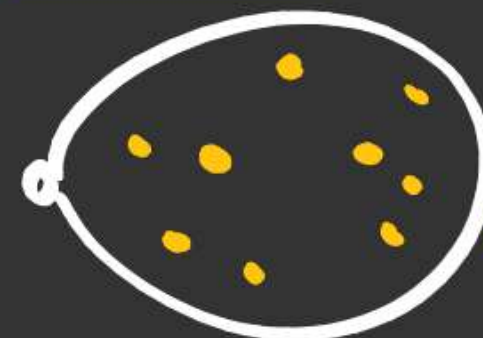
2 Liquid



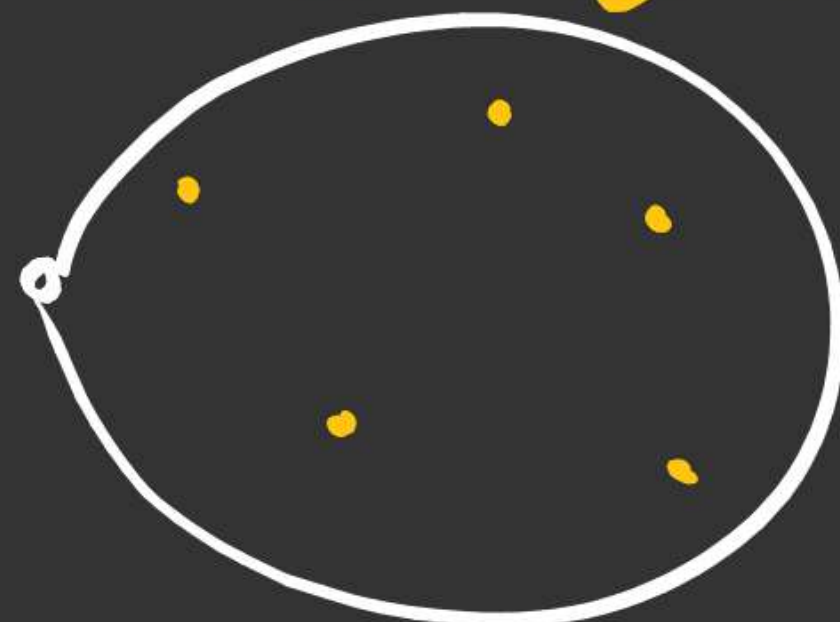
Temp

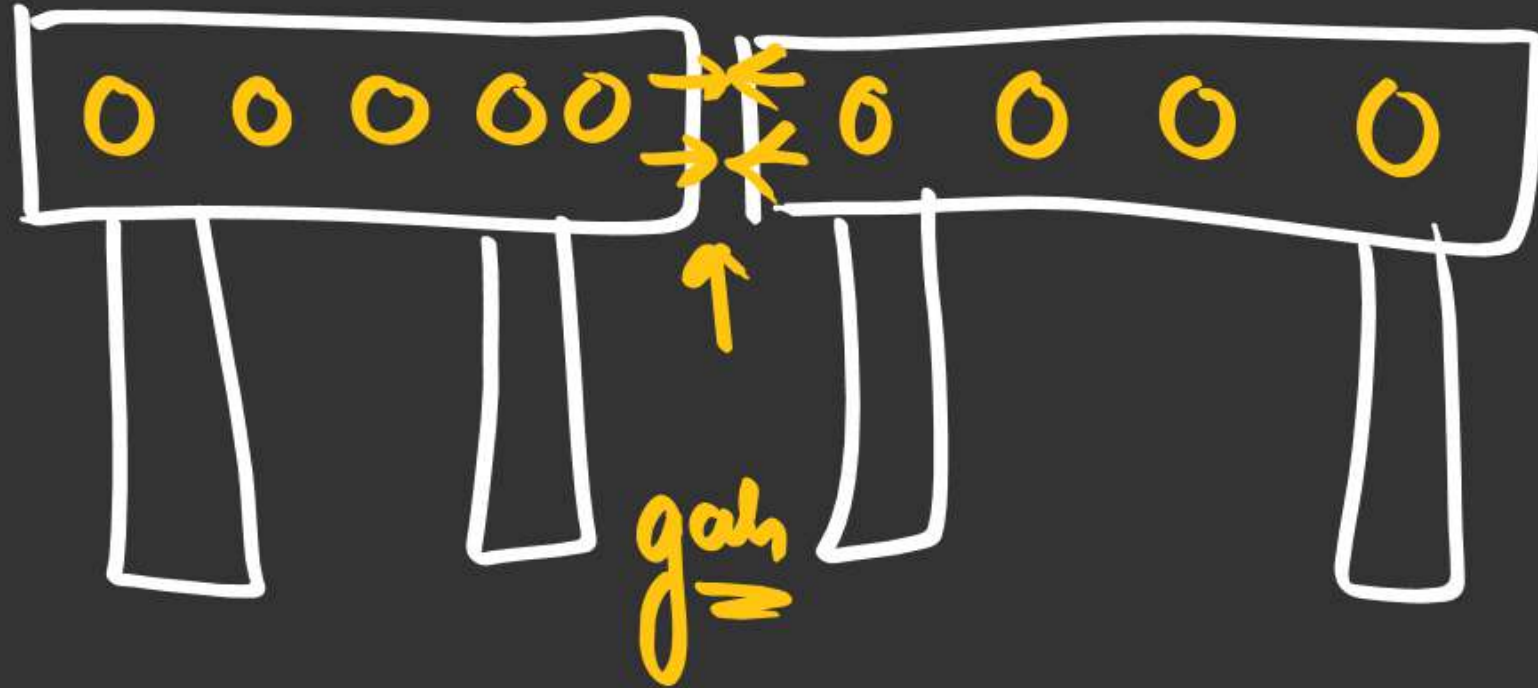


3 Gas



Temp
↑↑↑





$\Rightarrow$ Linear Expansion / α (coefficient of linear Expansion)

- let us consider the Rod of ^{math} initial length (l_0) when you give the temp.

the length of Rod **increase**

(1) $\Rightarrow l_0 < l$

This is **linear**

Expansion



$\Delta l = l - l_0$

change

$\Delta T = T_f - T_i$

Δl $\propto l_0$ — ①

Δl $\propto \Delta T$ — ②

on combining ① and ②

$\Delta l \propto l_0 \Delta T$

$\Delta l = \alpha l_0 \Delta T$ (coefficient of linear Expansion)

$l - l_0 = \alpha l_0 \Delta T$

$l = l_0 + \alpha l_0 \Delta T$



$$l = \underline{\lambda_0}^{\times 1} + \alpha \underline{\lambda_0} \Delta T$$

$$l = \lambda_0 (1 + \alpha \Delta T)$$

⑨

Superficial Expansion / Areal Expansion ($\beta \rightarrow$ coefficient of superficial expansion)

let us consider the Area $[A_0]$ if you increase the temperature of that Area it Expand this is known as Areal Expansion / Superficial Expansion



A_0
 T_i

10°C

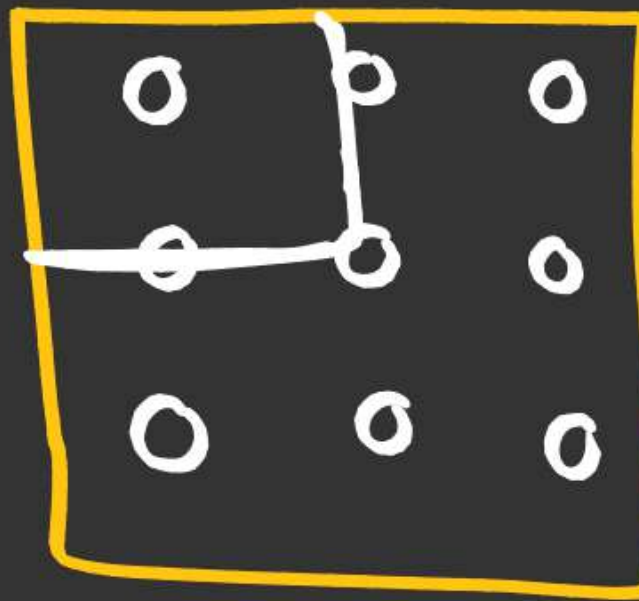
Maths

$$\Delta A = A - A_0$$

$$\Delta A \propto A_0 \quad \text{--- (I)}$$

$$\Delta A \propto \Delta T \quad \text{--- (II)}$$

on combining (I) and (II)



A

T_f 20°C

$$\Delta A \propto A_0 \Delta T$$

$$\Delta A = \beta A_0 \Delta T$$

coefficient of superficial expansion

$$A - A_0 = \beta A_0 \Delta T$$

$$A = \underline{A_0} + \underline{\beta A_0 \Delta T}$$

$$\Delta T = T_f - T_i$$

Superficial Expansion

$$A = A_0^{x_1} + \beta A_0 \Delta T$$

$$A = A_0 (1 + \beta \Delta T)$$

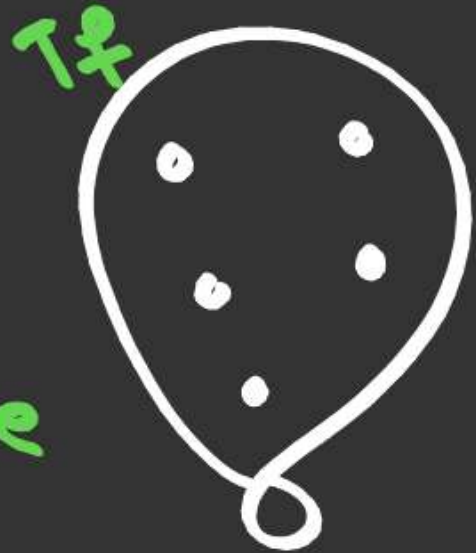
ii

Volume Expansion / Coefficient of volume Expansion

12

- let us consider the initial volume (V_0) on increasing the temperature

the volume increase this is known as Volume Expansion



$$\Delta T = T_f - T_i$$

$$\Delta V = V - V_0$$

math

$$\Delta V \propto V_0 \text{ --- (I)}$$

$$\Delta V \propto \Delta T \text{ --- (II)}$$

on combining (I) and (II)

$$\Delta V \propto V_0 \Delta T$$

$$\Delta V = \gamma V_0 \Delta T$$

$$V - V_0 = \gamma V_0 \Delta T$$

$$V = V_0 + \gamma V_0 \Delta T$$

Coefficient of Volume Expansion

$\propto$

$$\underline{V} = \underline{V_0^{\times 1}} + \gamma V_0 \Delta T$$

$$V = V_0 (1 + \gamma \Delta T)$$

13

Convection and Radiation

- **Convection:** Transfer of heat in liquids and gases by actual movement of particles.
- **Radiation:** Transfer of heat in the form of electromagnetic waves without any medium.



Thermal Conductivity

The ability of a material to conduct heat. It is defined as the amount of heat conducted per unit time through a unit area and unit thickness for a unit temperature difference.



Qualitative Ideas of Blackbody Radiation

An ideal body that absorbs all incident radiation and emits radiation of all wavelengths when heated is called a blackbody.



Wien's Displacement Law

The wavelength at which a blackbody emits maximum radiation is inversely proportional to its absolute temperature:

$$\lambda_{\text{max}} T = \text{constant}$$



Stefan's Law (Stefan–Boltzmann Law)

The total radiant energy emitted per unit area per unit time by a blackbody is directly proportional to the fourth power of its absolute temperature:

$$E = \sigma T^4$$

where σ is the Stefan–Boltzmann constant.



HOMEWORK



3 derivation (p)

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



Join Our Official Telegram & Whatsapp Channel



JKBOSE Wallah
Class 9th & 10th



JKBOSE Wallah



Scan the QR to Join

Thank
You

